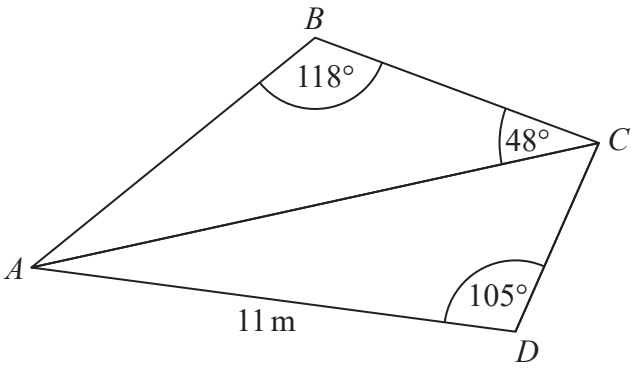
17 *ABC* and *ADC* are triangles.



The area of triangle *ADC* is 56 m^2

Work out the length of *AB*.
Give your answer correct to 1 decimal place.

In triangle ADC, use Area = ½ab sin C:

$56 = ½ \times 11 \times CD \times \sin 105°$

1 mark
Use the given area and included angle

$CD = 56 \div (½ \times 11 \times \sin 105°) = 10.54099384...$ m

1 mark
Find CD; retain the unrounded value

Cosine rule: $AC^2 = 11^2 + CD^2 - 2 \times 11 \times CD \times \cos 105°$

$AC = 17.09190364...$ m

1 mark
Find the shared side AC

In triangle ABC, use the sine rule:

$$\frac{AB}{\sin 48°} = \frac{AC}{\sin 118°}$$

$AB = AC \times \sin 48° \div \sin 118° = 14.38563268...$ m

1 mark
Pair each side with its opposite angle

AB = 14.4 m, correct to 1 decimal place.

1 mark
Correct final rounding

14.4 m

(Total for Question 17 is 5 marks)